

Lab 3 – Architecture Justification

Blood Bank Inventory & Emergency Donor Matcher

Architecture Selection:

We chose Layered Architecture for the Blood Bank Inventory & Emergency Donor Matcher. The system is organized into a Presentation Layer for the Medical Centre and Blood Bank Manager interfaces, a Business Layer for emergency request handling, inventory management, compatibility and donor matching, and notifications, and a Data Layer for persistent inventory, donor, and emergency-request data.

1) Clear separation of responsibilities:

The system contains safety-critical but distinct responsibilities: verified medical centres submit emergency requests, blood bank managers maintain inventory, compatibility rules determine suitable donors, and an SMS notification function alerts eligible donors. A layered design keeps the user interface separate from business rules and data storage, making each responsibility easier to understand, test, maintain, and modify without unnecessarily affecting the others.

2) Better fit for consistent blood inventory management:

The inventory ledger must remain accurate and must not allow the same physical blood unit to be allocated twice. Keeping inventory and matching logic within a coordinated business layer with controlled access to a shared data repository makes transactional consistency easier to enforce than a more distributed microservices design, which would introduce additional network and data-consistency complexity for this scope.

3) Security Advantage:

Sensitive donor information such as blood group, contact number, and location is not accessed directly by the presentation layer. Instead, access passes through business-layer components before reaching the data repository. This creates clear control points where authorization, input validation, TLS-protected communication, and encryption of stored donor data can be enforced, reducing direct exposure of personal and medical information.

4) Performance Benefit:

Emergency blood matching is time-sensitive. A layered design for this system can keep request validation, inventory lookup, compatibility checking, and donor matching within one coordinated application boundary, avoiding the extra network hops and service-to-service latency of a microservices deployment. This supports fast inventory updates and rapid matching before compatible-donor notifications are triggered.

Conclusion:

Layered Architecture provides a simple, maintainable, secure, and performance-conscious structure that fits the scale and safety requirements of the proposed blood-bank system.